

Threat Intelligence

Analyst Simulation — Capstone Project

Format
Individual submission

Est. Time
3–4 hours

Deliverables
Report + evidence

Level
SOC Analyst Tier-1

SCENARIO

You are a **Tier-1 SOC analyst**. Your SIEM has fired an alarm: workstation **WS-FINANCE-04** made repeated outbound connections to **185.220.101.47** on port 443 at 02:14 AM — outside business hours. The session lasted **18 minutes** and transferred **4.2 MB of data outbound**. No user was logged in at the time.

Your task: investigate this IOC using threat intelligence tools and frameworks, document your findings, and recommend a response.

PROJECT TASKS

Task 1 **Network IOC**

IOC Lookup & Classification

Look up the IP 185.220.101.47 in AbuseIPDB and AlienVault OTX. Identify and document the following:

- What threat category the IP belongs to (e.g., C2, Tor exit node, scanning, etc.)
- The confidence/abuse score from each platform
- Any known malware families or campaigns associated with it
- Which IOC type this is (Network, Host-Based, or Email) and why it falls in that category

Deliverable: Screenshot of lookup results from both platforms + 1 paragraph written analysis.

Tip: Use: <https://abuseipdb.com> and <https://otx.alienvault.com>

Task 2 **ATT&CK**

MITRE ATT&CK Mapping

Based on the observed behaviour (off-hours connection, large data transfer outbound, no logged-in user), map at least 3 MITRE ATT&CK techniques that could explain this activity.

- Technique ID (e.g., T1071.001)
- Technique name
- One-sentence justification explaining why it applies to this scenario

Deliverable: A table with columns: *Tactic* | *Technique ID* | *Technique Name* | *Justification*.

Tip: Reference: <https://attack.mitre.org>

Task 3 **TI Fundamentals**

Intel Type & Pyramid of Pain Analysis

Apply the concepts from the training to classify this investigation:

- Which of the 4 intelligence types (Strategic, Tactical, Operational, Technical) does this IP IOC represent? Justify your answer.
- Place this IOC on the Pyramid of Pain. What level does an IP address sit at?
- Explain what it means for the attacker if your team blocks this IP — is it a significant disruption or a minor inconvenience? Why?
- What higher-level IOC (from the Pyramid) would be more impactful to block?

Deliverable: Written response, 150–200 words.

Task 4 **Final Deliverable**

Incident Summary Report

Write a 1-page analyst report (300–400 words) addressed to your SOC Manager. The report must include all of the following sections:

- Executive Summary (2–3 sentences): What happened, what was found, recommended action.
- IOC Findings: IP address, abuse score, threat category, associated campaigns.
- ATT&CK Mapping: List the techniques identified in Task 2 in a compact table.
- Containment Actions: At least 2 immediate steps your team should take right now.
- Long-Term Improvement: One specific detection engineering or TI integration recommendation to prevent recurrence.

Deliverable: Polished written report, professional tone, 300–400 words.

Tip: This is the document you would hand to your manager. Write accordingly.

WHAT TO SUBMIT

#	Item	Format
1 — IOC Lookup	Screenshots from AbuseIPDB and AlienVault OTX + written analysis paragraph	Screenshots + text
2 — ATT&CK Table	Table mapping at least 3 techniques with Tactic, ID, Name, and Justification columns	Table (Word/PDF)

3 — Pyramid Analysis	Written response classifying the IOC type and Pyramid of Pain level	Word/PDF
4 — SIEM Rule	Pseudo-code correlation rule and TI feed integration description	Word/PDF
5 — Analyst Report	Full 300–400 word incident summary report addressed to SOC Manager	Word/PDF

Stay curious. Stay vigilant. Keep hunting.

This project is an individual assessment. All findings must be your own. You may use any publicly available TI tools and resources referenced during the training.